

Program : Diploma in Chemical Engineering	
Course Code : 4073	Course Title: Mass Transfer Operations I
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To enable the students to apply the principles of absorption, adsorption, drying and crystallisation in process plant operations.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I Mathematics II	1 2
Knowledge of basic chemistry		Applied Chemistry Applied Chemistry II	1 2
Knowledge of unit operations and processes in chemical engineering		Chemical Process Principles	2

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the principles of adsorption.	13	Understanding
CO2	Interpret the principles of absorption.	18	Applying
CO3	Interpret drying operation.	18	Applying
CO4	Interpret the principles of crystallization	11	Understanding

CO – PO Mapping with values

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2						
CO2	3						
CO3	3						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the principles of adsorption.		
M1.01	Summarise the principles of mass transfer.	3	Understanding
M1.02	Classify adsorption.	2	Understanding
M1.03	Summarize the properties, types and applications of adsorbents.	4	Understanding
M1.04	Outline adsorbent regeneration and adsorption equipment.	4	Understanding
Contents: Introduction to Mass Transfer Operations- Diffusion- Classification - Eddy diffusion- Molecular diffusion - Thermal diffusion - Molar flux- Fick 's law Adsorption - Physical and chemical adsorption Properties of adsorbents - Various types of adsorbents - Applications – Molecular Sieves - Ion exchange Adsorbent regeneration – Temperature swing adsorption, Pressure swing adsorption - Adsorption equipment – Constructional details and working of Fixed bed adsorber, Moving bed adsorber.			
CO2	Interpret the principles of absorption.		
M2.01	Summarise the principles of absorption and stripping.	3	Understanding
M2.02	Apply the theories of mass transfer in gas absorption.	7	Applying

M2.03	Illustrate absorption equipment.	6	Understanding
M2.04	Identify the difference in performances of packed tower and plate tower.	1	Understanding
	Series Test – I	1	

Contents:

Absorption and stripping - Applications - Henry's law - Mechanism of absorption operation - Condition of equilibrium between gas and liquid - Selection criteria for solvent in gas absorption

Material balance for an absorption tower - Simple problems

Absorption column –Plate column , Packed tower - Packing materials - Types and characteristics of packing - Regular and random packing - Channeling – Entrainment – Demister - Types of liquid distributors

Comparison of performance of packed and plate towers.

CO3	Interpret drying operation.		
M3.01	Outline drying operation.	3	Understanding
M3.02	Illustrate drying curves.	4	Understanding
M3.03	Apply the theories of drying to compute time of drying	6	Applying
M3.04	Outline the constructional details and working of different type of driers.	5	Understanding

Contents:

Drying - Principle - Industrial applications - Factors influencing the rate of drying - Equilibrium moisture content - Free moisture content - Critical moisture content- Bound water - Unbound water

Drying curves- Constant rate period - Falling rate period - Derivations of equations for time of drying in constant rate period and falling rate period

Simple problems to compute the time required for drying.

Classification of drying equipment - Constructional details, working and application of Tray dryer, Tunnel dryer, Rotary dryer, Spray dryer, Drum dryer, Fluidised bed dryer - Freeze drying.

CO4	Interpret the principles of crystallization		
M4.01	Outline crystallisation.	3	Understanding
M4.02	Identify the difference between evaporation and crystallization.	1	Understanding
M4.03	Summarize the methods of super saturation	2	Understanding

M 4.04	Outline the constructional details and working of crystallization equipment.	4	Understanding
	Series Test – II	1	
Contents: Crystallization - Definition – Principle – Applications Differentiate evaporation and crystallization Equilibrium curve for a solid-liquid system - Yield of crystals – Solubility – Saturation - Super saturation - Methods of super saturation. Constructional details and working of Agitated tank Crystallizer, Cooling crystallizer, Evaporative crystallizer, Swenson Walker Crystallizer.			

Text / Reference

T/R	Book Title/Author
T1	Treybal.R.E - Mass Transfer Operations
R1	K A Gavhane - Unit Operations. II
R2	McCabe and Smith - Unit Operations
R3	Surya Narayana - Mass Transfer Operations
R4	P.Chathopadhya - Unit Operations of Chemical Engg. Vol. – I

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/103/103103145/